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1 DECIPHERING BRAIN WITH SEMI-VR COGNITIVE GAMES

Your brain controls your body—with this basic concept, a Bengaluru based startup called Xeuron.ai is developing active intelligent footwear and virtual reality cognitive games as a solution for urban burnout and disease management



A lady got diabetes in her mid-30s and by the time she reached her early 40s, it started affecting her retina, teeth, and slowly each part of her body. This prompted her bioengineer son, Siddharth Nair, to innovate a non-invasive neural modulation system, which helped her walk again.

From here started a journey to develop intelligent connected systems to monitor and predict the risk of neurodegeneration. Siddharth established Xeuron.ai in mid-2019 but started working on it seriously only after the pandemic had hit.

The founder says, “The initial idea was to just develop smart footwear and insoles consisting of embedded sensors and then do a lot of things. But then, with all the emerging technology and the need

for adaptability to the user, it all came to a point of business sustainability. Hence, we migrated to semi-immersive virtual reality (semi-VR) to start with, and the smart footwear that we are developing would help build the whole ecosystem.”

Xeuron.ai works towards a hybrid approach for keeping the mind of a modern-age person young. It does so with the help of semi-VR games, where the body remains active while playing. Instead of just pressing the joystick, you actually run, jump, and lift yourself everywhere. Now, if you’re wearing smart footwear, it will look into all your bodily parameters.

Siddharth remarks, “While they are immersed in the game, we are able to track their body and mind such that we could help in prevent-

ing diseases. If they already have a disease, we could monitor how the disease is progressing. If they’re undergoing any treatment, we could look into the efficacy of the treatment. In future, we are hoping that we could also predict the disease.”

In 2020, the startup began developing smart footwear that would sense motions, pressure, temperature, and even the neuromuscular response of the person. “We got a couple of grants from DST-NIDHI’s PRomoting and Accelerating Young and ASpiring technology entrepreneurs (PRAYAS), IISc MSME, COE Clinic, and some prize money here and there,” informs Siddharth.

But when the pandemic initiated major disruptions in the semiconductor industry, Xeuron.ai decided to bring computer vision into the picture and build an AI around it. “I’m glad that we did that because, if you see, the scalability of a business is much faster when there’s less hardware to start with. And, of course, we can keep adding hardware later. So, we took that approach and hence we reached our MVP after a couple of pivots,” says the founder.

According to him, deciphering the brain and cognition is their biggest USP as no one is significantly working in this direction. “There